

EXTENDING A PHYSICS-BASED RECRYSTALLIZATION MODEL USING GENETIC PROGRAMMING

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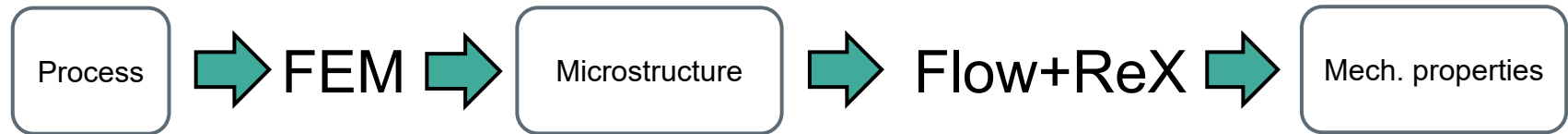
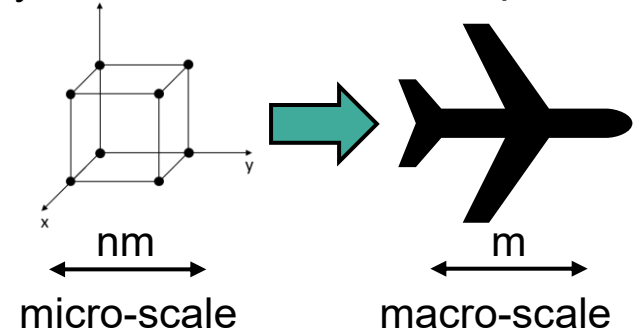


INTRODUCTION

Why microstructure prediction?

- Final grain structure affected by **extrusion parameters**.
- **Grain structure** controls **mechanical properties**.
- **Experimental** optimization is **expensive**.
- **Predictive models** needed for digital twins and process optimization:
 - **Physics-based** models provide **extrapolation** capability.
 - But: **calibration parameters** often have **unknown process dependencies**.

Crystal structure → Finished product



PREVIOUS WORK: SIMULATION WORKFLOW

Metal extrusion simulation workflow: FEM process model

- Aluminum billet is **pushed through** a ruler-shaped **die**.
- **Temperatures, strains, strain rates** are exported.

Streamline extraction workflow

- Process parameters **extracted from FEM model streamlines**.
- **Grain size evolution** by parallel microstructure calculations.

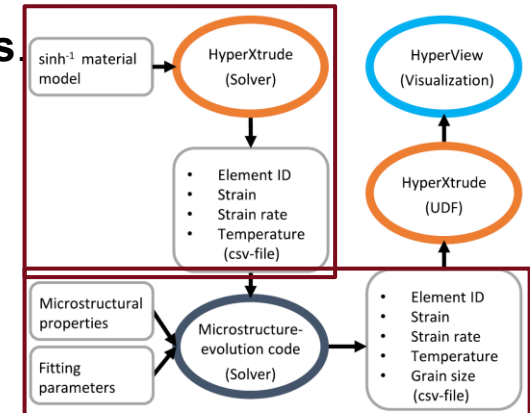
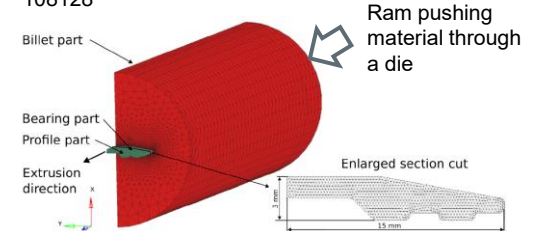
Physics-based microstructure model

Tailored for 6xxx-series aluminum alloys. Including:

- **deformation history** (dislocation density-based flow stress)
- **grain morphology** (dynamic and static recrystallization)



Hovden, S. et. al. Parameter study of extrusion simulation and grain structure prediction for 6xxx alloys with varied Fe content, 2024-03, Materials Today Communications, Vol. 38, Elsevier BV, p. 108128



PREVIOUS WORK: FLOW STRESS MODEL SR

Basic workflow:

- Compression tests (depending on temperature and strain rate).
- Physics-based MD²M (mean dislocation density dependent flow model).
- Parameter fitting of alloy dependent coefficients $\rightarrow A, B, C$
- Flow stress model:
 - Current local flow stress: $\sigma = f(\rho)$
 - Dislocation density evolution (time, T ...temperature, ϕ ...strain):

$$\Delta \rho^t = \rho^{t-1} + \left(\frac{1}{\rho^{t-1}} - B \cdot \rho^{t-1} \right) \phi - \frac{C}{T} [(\rho^{t-1})^2 - \rho_{eq}^2] \cdot \Delta t$$

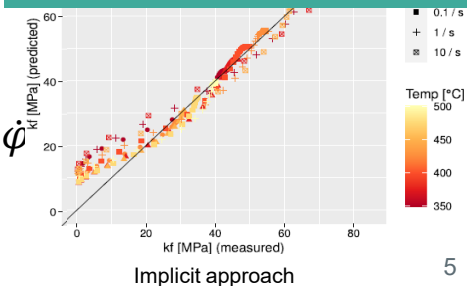
Novel workflow:

- Statistical regression to find $A = f(T, \phi)$, $B = f(T, \phi)$, $C = f(T, \phi)$
- e.g.: $A(T, \phi) = (0.11 \cdot T + 3.7 \phi - 17.3) - 0.069 \log(\phi)$

Kronberger, Kablman, Kronsteiner, Kommenda,
Extending a physics-based constitutive model using
genetic programming, 2022-03, Applications in
Engineering Science, Vol. 9, Elsevier BV, p. 100080



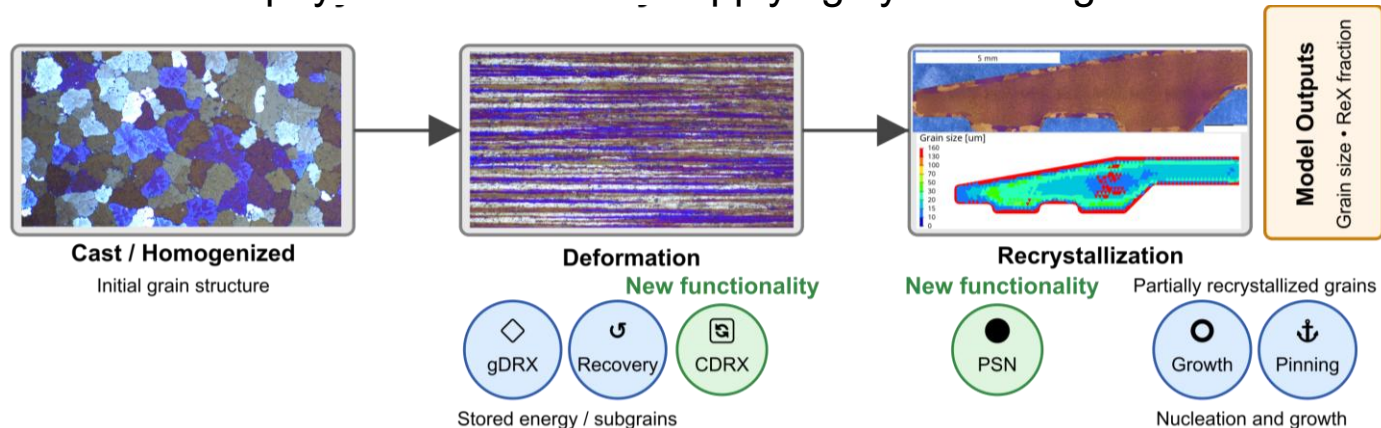
- Physics-based model retained.
- Calibration parameters became process-dependent.
- Improved interpolation between test conditions.
- Interpretable alternative to black-box models.



MOTIVATION: CAN WE TRANSFER THIS TO RECRYSTALLIZATION?

Same concept: replace uncertain calibration parameters with symbolic equations. But:

- ReX models contain many uncertain parameters and **missing physical mechanisms**:
 - Continuous Dynamic Recrystallization (**CDRX**)
 - Particle Stimulated Nucleation (**PSN**)
- Need to extend ReX physics **FIRST** before applying symbolic regression:



TRANSFERRING SYMBOLIC REGRESSION TO RECRYSTALLIZATION

Experiments (compression, annealing)



Physics-based ReX Model (fortran code)



Parameter Identification (regression methods)



Implementation of ReX Model in Julia



Symbolic Regression (Genetic Programming)



Hybrid ReX Model: Calibration parameters →
process-dependent equations

Previous Work

Physics-Based Flow Model (MD²M)

Calibration Parameters A, B, C

Symbolic Regression / Genetic Programming

$$A = f(T, \epsilon)$$

$$B = g(T, \epsilon)$$

$$C = h(T, \epsilon)$$

Current Work

Improve AA6005A Recrystallization Model

+ Continuous Dynamic Recrystallization (CDRX)

+ Particle Stimulated Nucleation (PSN)

+ Improved Static Recrystallization

+ Implementation of ReX model in Julia

Future Work

Apply GP / SR to Recrystallization

Learn Parameter Dependencies

CDRX • PSN • Growth Kinetics

Hybrid Physics + Data-Driven ReX Model

SIMULATION METHODS AND IMPLEMENTATION

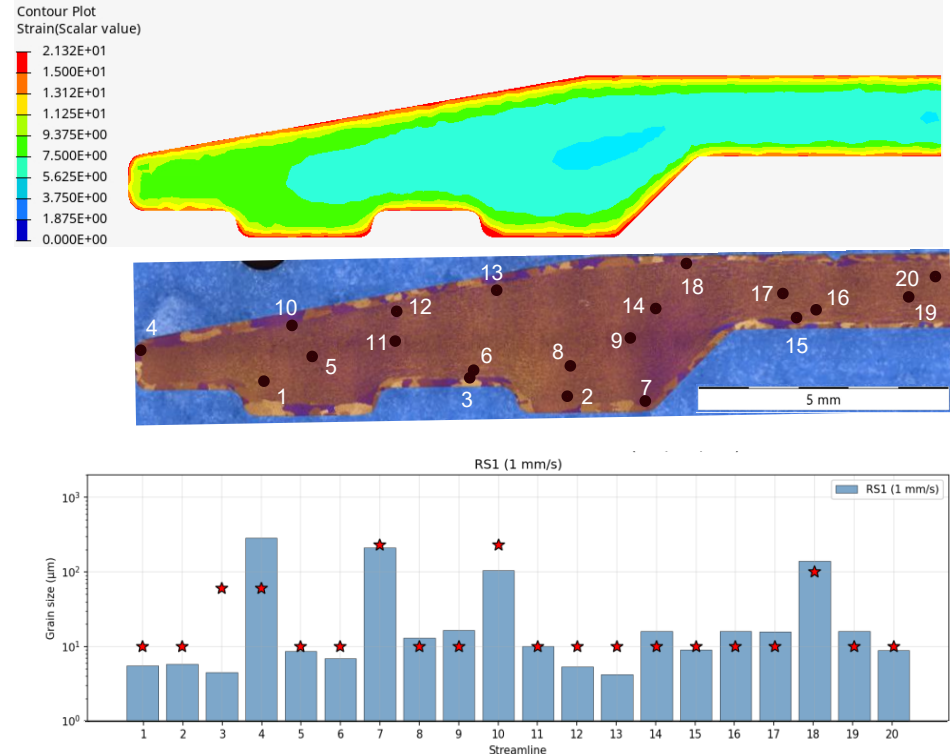
Process-dependent microstructure properties:

- **Temperature, strain and strain rate**-dependent flow stress model based on a Kocks-Mecking (KM).
- **Dynamic and static recrystallization** model based on a Johnson-Mehl-Avrami-Kolmogorov (JMAK) and Kocks-Mecking approach.
- Grain morphology is considered by:
 - **Geometric dynamic recrystallization** (gDRX): modelling breakage during forming.
 - **Continuous dynamic recrystallization** (cDRX): lattice rotations at large deformations.
 - Static recrystallization: **Grain Boundary and Particle-stimulated Nucleation** (PSN).

RESULTS EXTENDED REX MODEL

Ram speed 1 mm/s:

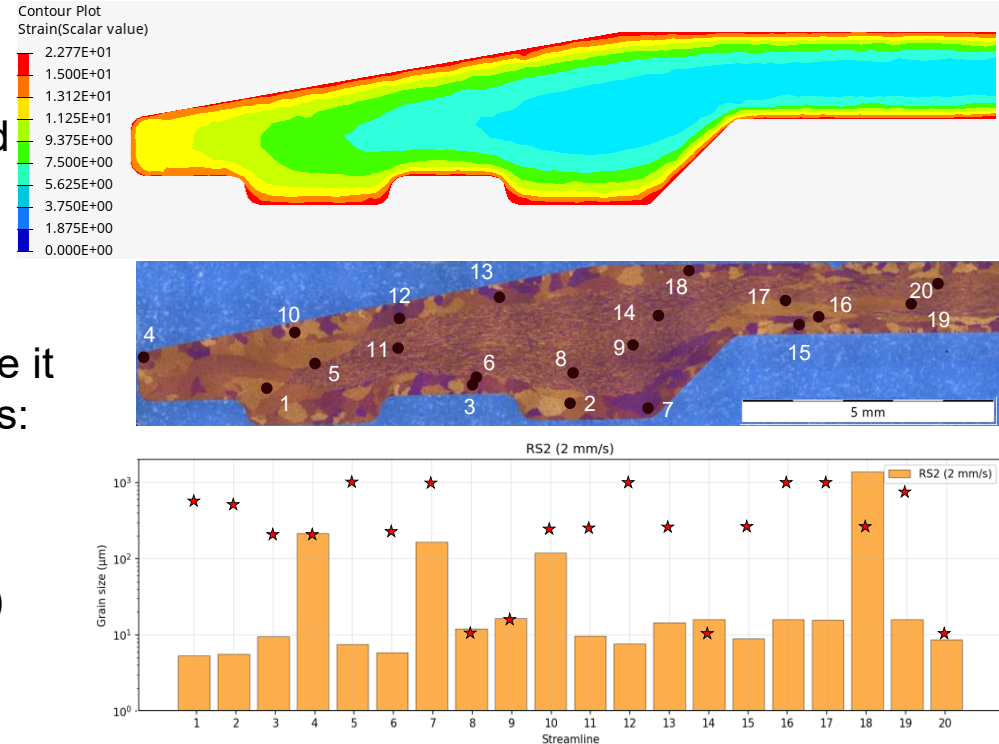
- Mostly **good agreement** with measured results.
- **Interpretation** of the grain sizes from micrographs is **difficult**/not precise.
- Extraction of streamlines allows free positioning but elemental and nodal results are interpolated → **sharp transition** from large to small grains → **high uncertainty**.
- Small grains in the micrograph are supposedly **large needle-shaped grains**.



RESULTS EXTENDED REX MODEL

Ram speed 2 mm/s:

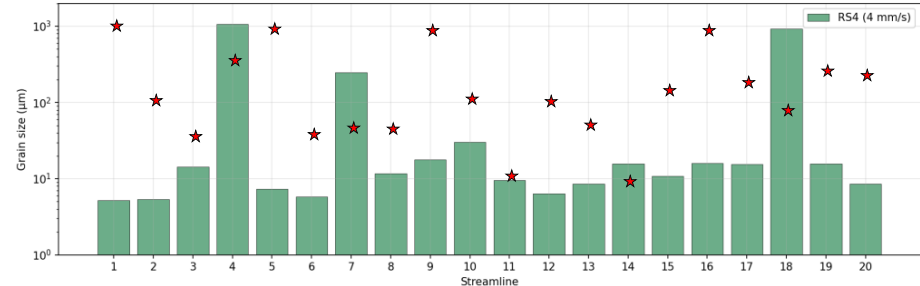
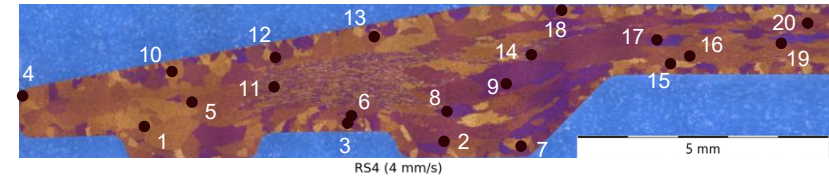
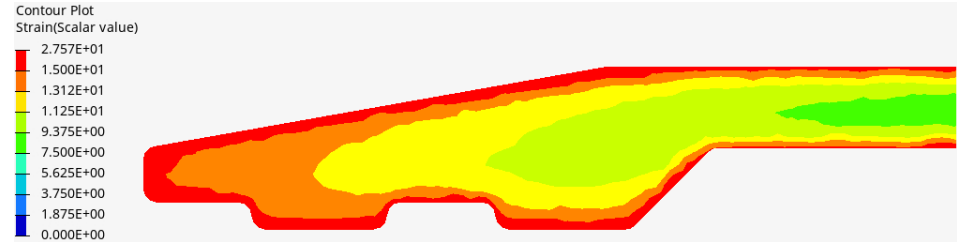
- **Problem to recreate experimental measurements** even with extended physically-based model.
- **Higher strain values for higher ram speeds.**
- But: **competing mechanisms** make it **hard to calibrate** model parameters:
 - CDRX nucleation coefficient
 - PSN coefficient
 - Grain boundary mobility (growth)
 - Critical strain parameters (gDRX/CDRX transition), ...



RESULTS EXTENDED REX MODEL

Ram speed 4 mm/s:

- Representation of grain sizes only partially by strain.
- Consistently larger grains only at the surface.
- The **influence of higher strains** on grain size is **not properly reproduced**.
- **Use symbolic regression** to enhance the model for better reproduction of experiments.



CONCLUSIONS AND OUTLOOK

- Symbolic regression successfully applied to flow-stress model (MD²M)
 - Unknown parameters A, B, C → interpretable equations $f(T, \dot{\phi})$.
 - Physics-based model structure fully retained.
- Recrystallization model significantly extended:
 - CDRX, PSN, improved SRX, growth, nucleation framework.
 - Validated for AA6005A — improved microstructure prediction.
- Foundation laid for symbolic-regression-enhanced ReX modelling:
 - Next step: apply GP to ReX calibration parameters.

THANK YOU!

Johannes Kronsteiner, July 20th , 2026

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